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Measuring perceived difficulty in video games: Development of the subjective game difficulty scale

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ABSTRACT

Game difficulty is a significant focus within game design research. However, how to effectively measure players' perceived difficulty (subjective game difficulty, SGD) remains unresolved. Currently, SGD is primarily assessed by the player's difficulty rating, self-reporting, and physiological measurements. However, these measuring methods are limited in their ability to capture the complex structure of SGD for complete and precise evaluation. Therefore, this study develops a new scale to measure SGD. We first identified and classified the structure of SGD in six dimensions. On this basis, the Subjective Game Difficulty Scale (SGDS) was developed and validated through a standard three-stage scale development method. Sixty related items were generated, and thirty-three items were selected in the first two stages. In the third stage, an international survey with 326 American (USA), Chinese, and Japanese participants was conducted to test the scale. The results indicated that our final 25-item SGDS is reliable and valid. We discussed the usability of the SGDS, compared it with other measurement methods, and we provided design implications and the guidance regarding how to apply the SGDS. This work presents a promising instrument for game designers and researchers to support game difficulty evaluation and design.

1. Introduction

Game difficulty (or game challenge) is one of the essential components of gameplay in video games [1] and impacts player–game interaction [2] and player experience [3]. Therefore, game difficulty design, as a critical part of game design, is of great interest to designers and researchers. Designers tend to believe that well-designed game difficulty is balanced and can encourage players to be engaged and immersed, thus gaining an excellent experience [4–6]. However, players vary in their levels of game skills, and game skills tend to improve during the playing process. Therefore, aiming to design the optimal game difficulty requires designers to measure game difficulty more precisely to evaluate their design. However, the current measurements of game difficulty vary in their assessment of objects and effects.

The terms "challenge" and "difficulty" are often used interchangeably in game research [7], which compromises the attempts to construct measurable difficulty. Although studies applied the challenge concept to develop the measuring instruments (e.g., [8,9]), we suggest that the concept of game difficulty is more solid than game challenge in measuring usage. In research, the term challenge is more polysemous (variable in meaning) and is to be regarded as a nontrivial task [1], a kind of player experience [8], or a type of motivation [10]. This polysemy of challenge makes the measured objects subtle and uncertain (e.g., measuring the challenge of tasks or experience), which does not meet the measurement requirements for precision. In contrast, the game difficulty conceptually refers to the level of demand the game imposes on player skills [11–13]. This definition identifies the object measured as this demanding level, which is practical for measurement. In addition, this demanding level can be assessed through objective and subjective aspects, which provide a more precise scope for the measurement.

Therefore, researchers have recently divided game difficulty into subjective game difficulty (SGD) and objective game difficulty (OGD) for measurement [14–16]. Specifically, OGD denotes how the game's complexity causes a certain level of demand on players, which can be measured by the attributes of game tasks like task-demand speed, time, etc., [17,18] or a player's failure probability during the game [19]. In comparison, SGD is well known as perceived difficulty, which means how the player feels and evaluates such a demand from their subjective experience in the game [8,20]. Currently, SGD is measured through three main approaches: simple self-report (questions) [21], structured self-report (scales) [8], and physiological measures [22,23].

It is reasonable that OGD and SGD are generally thought to match since they are two sides of the game difficulty. However, researchers

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found that OGD does not match SGD completely through current measuring methods, and some players tend to underestimate the nuances in difficulty (i.e., SGD lower than OGD) [15]. Other game design research has reported the complexity of players' SGD evaluation [24–26], which indicates that present SGD assessment methods are incomplete and therefore imprecise. Based on our literature review, we suggested that the complexity of SGD could be explained by its multidimensional structure. However, the present SGD measuring instruments can hardly encompass all the dimensions with no consideration of informative distinctions. As objective measuring instruments, physiological measures must first be based on self-report data as the interpretive material (e.g., [27]). For self-report methods, studies usually measure SGD through a few simple questions [25,28,29], while current established SGD scales are based on the challenge concept [8,9]. Therefore, these methods are not as precise for measuring the multidimensional SGD effectively.

This work aims to develop a new scale to measure SGD in video games. Firstly, a systematic literature review is conducted to determine the definition and structure of SGD and how to measure it. On this basis, we propose an analytical structure for SGD and summarize its six dimensions as follows: *game complexity*, *game completion difficulty*, *game-playing difficulty*, *player competence*, *player pressure*, and *player effort* [7–10,22,30–34] (Section 2). Subsequently, we developed the *Subjective Game Difficulty Scale* (SGDS) through the standard three-step method of item generation, scale development, and scale testing based on these six dimensions. Sixty related items were generated in the first stage based on the literature and existing scales, and thirty-three items were retained in the second stage through an online card-sorting investigation. In the third stage, we conducted an international online survey to test the scale. All subjects were recruited from crowdsourcing websites. A subject selection process was set to ensure that they were familiar with the game used in the testing. The final 25-item SGD scale was proven to have good reliability and validity (Section 3). In Section 4, we summarized the usability of the developed scale and discussed its differences with other SGD-measuring instruments. We also explained how to apply the developed scale and provided this study's implications for game design research.

Our work contributes by developing and validating a new scale of SGD. The six dimensions for constructing SGD were introduced to deepen the current understanding of game difficulty. The developed 25-item SGD scales in English, Chinese, and Japanese were provided. We expect the provided measuring instrument to support future game difficulty measurement and design evaluation for researchers and designers.

2. A systematic literature review on subjective game difficulty

We systematically reviewed the literature on subjective game difficulty following PRISMA guidelines [35], see Fig. 1. Specifically, we searched for the following terms: (game difficulty OR game challenge) AND (measuring OR measurement OR scale) in the ScienceDirect, ACM Digital Library, and IEEE Xplore Advanced Search, with the search area limited to Titles, Abstracts, and Keywords of the papers published in the past 10 years (from 2014 to 2024). A total of 2167 publications were found. These papers were selected independently by two reviewers. Any disagreements regarding paper selection were resolved by discussion with a third researcher. After excluding (1) non-English literature, (2) non-research articles, (3) non-video game papers, and (4) other papers less relevant to the game difficulty's measurement or evaluation topic, 243 papers were retained. We further screened papers to exclude (1) short papers and (2) papers with unclear or limited contributions, which include those using existing or similar approaches to measure difficulty or design difficulty algorithms. In this step, 201 papers were excluded, and 42 papers were retained. As supplements, we also investigated other topic-relevant papers from the references of the found papers (including those from other databases). Finally, 68 papers and books from our search were included and cited in this work.

Reviewing these papers, this section first investigates the concept of game difficulty and summarizes the definitions of objective game difficulty (OGD) and subjective game difficulty (SGD). Subsequently, we analyze the multidimensional structure of SGD evaluation and propose six dimensions for measurement. Finally, we introduce current SGD-measuring methods and their limitations.

2.1. Understanding the game difficulty concept

The concept of game difficulty originated from the concept of task difficulty in the fields of psychology, education, and human–computer interaction (HCI) [11–13]. Simply put, a task has a specific structure consisting of rules, goals, and other elements; this structure (i.e., objective task complexity [30]) makes it difficult to complete the task (i.e., failure is more likely). In these fields, thus, task difficulty can be quantified as the complexity of the task or the level of demand the task imposes on performers [11–13]. Games usually have typical task structures, so the difficulty of a game can also be regarded as the difficulty caused by the tasks in the game. Many game design studies directly use the complexity of game tasks to quantify and design game difficulty (e.g., scroll speed and target size in platformer games) [36–38]. However, this approach considers little about the impact of the human factor in game difficulty, especially the differences in players' game skills. As an improvement, Aponte et al. [19] proposed that game difficulty can be represented as a probability function of player failure at a specific time. This quantifying method of difficulty relates more to the evaluation and prediction of player performance (failure) and is applied in recent studies [39,40].

However, this way of defining and measuring game difficulty did not reflect the subjective feelings of difficulty of the players themselves [25]. Therefore, as an alternative, some research uses perceived difficulty to describe each player's subjective feelings of difficulty in the game [14,21,28]. Perceived difficulty is usually used in psychology research and refers to people's opinions about whether a behavior is perceived to be difficult to perform or not [41]. In game research, for example, Li et al. [14] proposed that game difficulty consisted of objective difficulty and player-perceived difficulty. They further argued that objective difficulty is mainly determined by the demands of the task, but perceived difficulty varies from person to person. Demediuk et al. [21] measured perceived difficulty by asking players how they felt about the difficulty of defeating the enemy in the game. Constant et al. [15] first divided game difficulty into objective difficulty and subjective difficulty, based on the study measuring game difficulty by Aponte et al. [25]. They proposed that subjective difficulty was a psychological construct of the player related to their evaluation of failure probability. We agree that game difficulty should be divided into these two parts, and that "subjective difficulty" better describes the generation of the evaluation-based feelings of difficulty than "perceived difficulty". However, it is narrow to regard subjective game difficulty (SGD) as an evaluation of failure probability (which may be called "failure expectation"). In line with Denisova et al. [8], we suggest that SGD should be complex and comprehensive feelings based on evaluating difficulty, i.e., a subjective experience rather than merely an expectation.

Many studies have shown that players tend to have complex evaluations of their game-playing to form SGD [24–26]. Research indicates that the complexity of SGD is related to the objects and patterns of the player's evaluation [8,42,43]. Since evaluation patterns are more related to the personality and preferences of players, studying how evaluated objects affect players' feelings of difficulty is valuable for understanding SGD. However, few studies focus on these objects, instead assessing players' overall feelings of difficulty through simple questions. Recent studies have attempted to clarify these evaluated objects by analyzing challenges. For example, Vahlo & Karhulahti [9] summarized

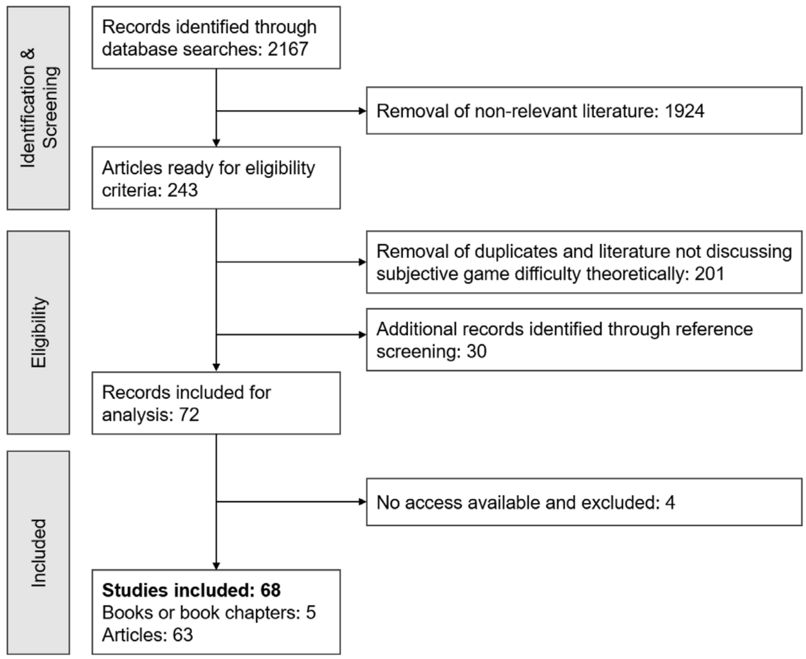

Fig. 1. We conducted a systematic review to identify SGD-related literature following PRISMA guidelines. Finally, 68 studies are included.

four core factors of challenges: physical, analytical, socioemotional, and insight. Using the term "perceived challenge" instead, Denisova et al. [8] suggested it was the player's experience of challenge posed by a particular game and proposed four categories: emotional, performative, cognitive, and decision-making challenges. The term "perceived challenge" addresses the problem of ambiguity in the challenge concept. In addition, the proposed challenge categories in these two studies explain several sources of player feelings of difficulty. However, these categories are still not comprehensive enough since such feelings are also affected by other evaluating factors, such as the pressure players feel in the game [44]. How players form SGD in their evaluation of difficulty-related objects remains unclear.

To identify how players make their SGD evaluations, it is necessary to consider how game difficulty emerges. The interaction perspective from HCI provides a valuable lens for understanding game difficulty [8]. The interaction process is mainly about how humans interact with computers through information input and output [45]. Specifically, the game-playing process can thus be regarded as the player–game interaction process in which players receive, understand, and complete game tasks through output and input devices [2]. Therefore, the game difficulty can be considered to occur during the interaction between the player and the game task. Current objective game difficulty (OGD) can be defined as *the objective level of demand the game task imposes on player skills during the interaction between the player and the game task*. SGD is more complicated, but the interactive perspective provides basic evaluation objects for players in forming SGD. Literature also supports the view that the player will evaluate the three elements in the interactive process (i.e., the game task, the game process, and the player's self-state) to generate subjective feelings of difficulty [14,31,46]; see Fig. 2. Therefore, this study defines SGD as *the player's subjective feelings of difficulty based on their structured evaluation of the game task, game-playing process, and their respective states*.

2.2. Multidimensional structure of SGD evaluation

SGD is derived from players' structured evaluation of game tasks, game-playing processes, and their states, but the structure of the evaluation remains unclear. Therefore, we reviewed the literature on these three interaction elements and summarized the six dimensions to build the evaluation structure.

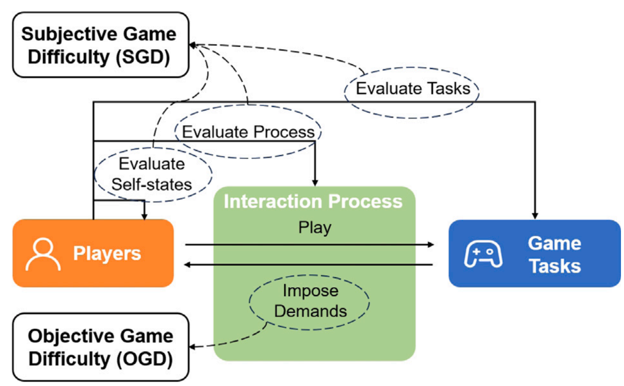

Fig. 2. How objective game difficulty (OGD) and subjective game difficulty (SGD) emerge in the player–game task interaction.

Regarding the game task element, we found that players tend to generate two aspects of views by evaluating it. One kind of view is based on the complexity of the game task, as introduced by the theory of Task Complexity [11]. According to Liu & Li [30], subjective task complexity is the task performer's feeling about the degree of complexity of the task. We agree that by observing the task elements (e.g., rules, goals, etc.), players would perceive the task as complex, or not complex. Another view is about the player's anticipation of the level of completion of the game task. Constant et al. [15] measured the SGD by each player's estimate of their likelihood of failure. We suggested that this failure/success expectation is based on a general evaluation of whether they can complete the game task. It can be considered that if a task is complex and hard to complete, it is difficult for players. To summarize, players will evaluate game tasks from both perspectives of elemental complexity and overall completion possibility to generate feelings of difficulty.

Regarding the interaction element, players will evaluate their game-playing process. In this process, the game task imposes demands on the player, and the player, with a specific skill level, expresses corresponding performance [8]. Therefore, players also evaluate two aspects of the interaction process, i.e., the demands arising from tasks and

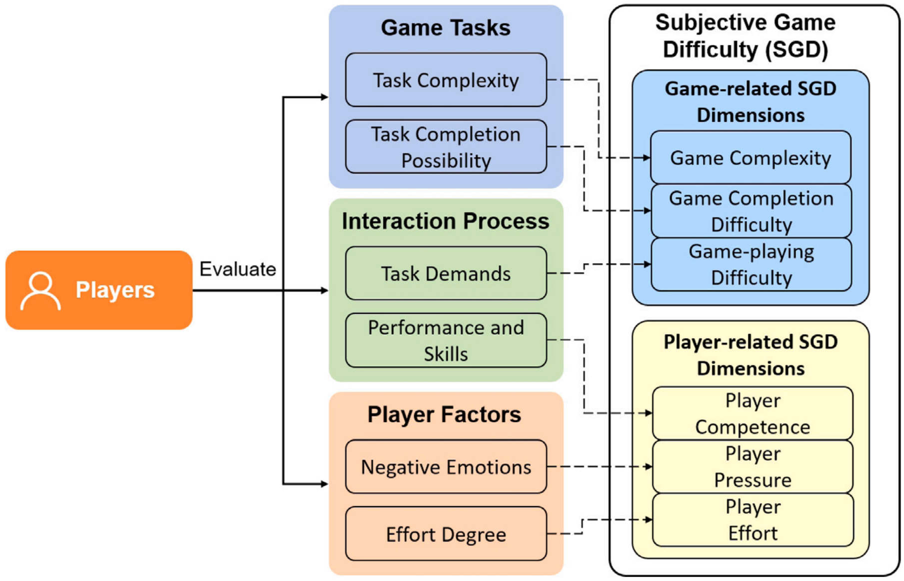

Fig. 3. We summarize the SGD structure and dimensions by analyzing how players evaluate the three elements of player–game interaction. SGD has two parts, game-related SGD and player-related SGD, and six dimensions: game complexity, game completion difficulty, game-playing difficulty, player competence, player pressure, and player effort.

their own performance and skills. In other words, player evaluations of OGD in interaction also generate feelings of difficulty. As we introduced in Section 2.1, researchers have determined different sources of challenges, such as physical, cognitive, etc. [8,9]. These types of challenges represent the various aspects of game task demands, and they affect players' subjective feelings of difficulty [11,47,48]. On the other hand, players' feelings about their performance and skills in specific tasks have been studied and named as player competence [10,32]. As introduced in Self-determination Theory [49], a player's sense of competence is about whether players believe they are skilled and can play well in the game or not. This perception partially overlaps with the concept of self-efficacy, which refers to task performers' opinions about their capabilities on specific tasks [30,50].

Considering the player element, players will conduct a self-evaluation to determine whether they feel the game task is difficult. According to the literature, the two player factors that are most related to SGD are the degree of subjective effort and negative emotions [30,34,51,52]. According to research in the fields of psychology and HCI, cognitive load (i.e., the mental demand imposed on the performer by the task [53,54]) was introduced in quantifying task difficulty [22,33,37,55]. Perceived effort and perceived pressure are commonly used to measure cognitive load through the self-report method [11,51,52,56]. In game research, player effort and frustration were also mentioned when describing the related feelings of difficulty or facing failure [7,34]. It is natural to assume that the harder the game, the more effort is needed and the more negative feelings players have [57]. Therefore, we believe players will evaluate their effort and negative emotions in playing the game to form feelings of difficulty.

Therefore, combining these views with the theories of Task Complexity [11], Self-determination [49], and Cognitive Load [54], we structured SGD into two parts and six dimensions. See Fig. 3. On the one hand, players will evaluate game task complexity, completion possibility, and demands to form the game-related part of SGD, and the corresponding three dimensions are game complexity, game completion difficulty, and game-playing difficulty. On the other hand, players will evaluate their performance and skills, negative emotions, and effort degree to form the player-related part of SGD, and the corresponding three dimensions are player competence, player pressure, and player effort. For the detailed definitions of each dimension, please see Appendix A.1. It should be clarified that besides the feeling of stress, the pressure dimension further includes the player's perception of other negative emotions related to pressure, such as anxiety and frustration.

2.3. Comparison of SGD-measuring methods

There are currently three main methods for measuring SGD: simple self-report through several questions, structured self-report through scales, and physiological measurements. We introduce these methods and compare their measurement scope and limitations; see Table 1.

In game research, a common method is to ask players to rate their perception of the difficulty of the game via a few questions, i.e., the simple self-report method [21,28,36]. Although this simple self-report method may help understand the player's overall feeling of difficulty, it is ineffective in providing details. As we mentioned in Section 2.2, players generate SGD by evaluating different objects. However, these objects are hard to identify by merely measuring an overall feeling of difficulty and thus, missing details on why players feel difficulty. For example, players may think the game is difficult because they have too many failures (player competence) or the game task looks too complex (game complexity).

The use of physiological measurements is still limited since there are no mature approaches for measuring SGD. In HCI and psychology, physiological techniques were used to measure workload [61,62], especially the effort and pressure of task performers. For example, indicators like heart rate, respiration, and blink rate were used to measure the objective state of investment/stress of performers [59]. Some game studies attempted to measure game difficulty through physiological indicators like EEG and eye movement [23,58,63]. AI technologies were also used to model the relationship between physiological signals and SGD [64–66]. However, physiological measurement methods are not complete and are not easy to use, so that evaluations still depend on self-reported SGD results to provide interpretability. Therefore, they cannot currently be used boldly. Nevertheless, physiological measurement methods have great potential in the future due to their objectivity, real-time feedback, and non-intrusiveness.

The scales that can be used in SGD measurement include the *Challenge Originating from Recent Gameplay Interaction Scale* (CORGIS) [8] and the *Videogame Challenge Inventory* (CHA) [9] from game research, and the *NASA-Task Load Index* (NASA-TL) [56] from HCI. In addition, some subscales of the *Player Experience of Need Satisfaction* (PENS) scale [60] and the *Intrinsic Motivation Inventory* (IMI) scale [10] can also be used. CORGIS and CHA are scales developed for game challenges, and both scales are constructed by analyzing the source of challenges.

Table 1
Comparison between different measuring methods of SGD: simple self-report, structured scales, and physiological measurement. CORGIS: Challenge Originating from Recent Gameplay Interaction Scale, CHA: Videogame Challenge Inventory, NASA-TL: NASA-Task Load Index, IMI: Intrinsic Motivation Inventory, PENS: Player Experience of Need Satisfaction.

Methods	Instruments	Measurement Scope	Limitations
Simple Self-report	Questions that ask players to rate the difficulty level [21,28,36]	The player's overall difficult feeling.	Missing details on why players feel difficulty.
Physiological Measurements	Heart rate, respiration, blink rate, eye movement, EEG, etc. [23,58,59]	The player's effort and negative emotions (e.g., pressure, anger, etc.).	This method is currently not completed and is not easy to use.
Structured Scales	CORGIS [8]	The player's perceived challenges of emotional, cognitive, performative, and decision-making; perceived effort.	Lacking the measurement of the subject's evaluation of task complexity and their pressure feelings.
	CHA [9]	The player's perceived challenges of physical, analytical, socioemotional, and insight.	For measuring the subject's challenge preferences rather than the level of challenges.
	NASA-TL [51]	The task performer's evaluation of task demands (mental, physical, and temporal) and their performance, effort, and frustration.	Lacking the measurement of the subject's evaluation of task complexity; each subject-related dimension has only one question.
	IMI [10]	The SGD-related subscales measure the subject's perceived competence, effort, and pressure.	Lacking the measurement of the subject's evaluation of task-related dimensions.
	PENS [60]	The SGD-related subscale measures the player's perceived competence.	Lacking the measurement of the subject's evaluation of task-related dimensions and their effort and pressure feelings.

However, CHA focuses more on measuring players' preferences for challenge types rather than the level of challenges. CORGIS provides four challenge categories that reflect different aspects of the task's demands on players. Additionally, some items in it also measure the perceived effort of players, such as "I invested much thought into the game" and "Playing the game requires great effort". However, the measurements of players' views on game tasks and their own stress in this scale are lacking. NASA-TL is widely used to measure the load in interaction. Its measurement scope covers task demands (divided into three categories: mental, physical, and temporal) and the performance, effort, and frustration of the task performer. However, this scale still lacks the measurements of task complexity, and the three dimensions related to the task performer of this scale include only one question each. Similarly, the IMI scale includes three dimensions related to task performers: competence, effort, and pressure, while the PENS scale only includes the player competence dimension, and neither scale includes task-related assessment dimensions.

In summary, current methods cover different aspects of SGD measurement but they are still not comprehensive and inclusive. Therefore, the next section introduces how we developed a new inclusive scale that measures the six dimensions of SGD.

3. The development of the subjective game difficulty scale

Based on the literature [67,68], the development of our scale was in three stages: (i) item generation, (ii) scale development, and (iii) scale testing. In the first stage, we created items for each dimension of SGD based on current measuring instruments and literature. The purpose of this stage was to create enough items for filtering in the scale development stage. In the second stage, we applied an item sorting procedure by recruiting expert game players, and the ineffective or ambiguous items were removed after analyzing the sorted results. In the final stage, we tested the instrument through an online survey.

3.1. Stage i: Item generation

We created the items for each dimension based on current scales and literature (see Tables 2 and 3). It should be clarified that we adapted some existing items from other scales and added them to our initial item pool. This method is feasible in scale development and makes full use of existing validated items [68]. We also noticed that the two dimensions of game complexity and game completion difficulty can be measured before playing, while the other dimensions can only be measured during or after the game. This is because players can form their perceptions of game complexity and completion probability based on observation of the game contents or by observing the play of others. However, this perception is achieved without actually playing the game, which is more of a first impression of the game and is likely to change after playing. In comparison, other dimensions measure players' feelings produced during and after the specific playing process and thus cannot be measured before this process. In addition, the player competence dimension asks the player, based on their playing, to evaluate their "present competence" of the game. Therefore, we utilized the present tense to create the items of the three dimensions: game complexity, game completion difficulty, and player competence. For the other three dimensions, i.e., game-playing difficulty, player pressure, and player effort, we used the past tense to create the items.

For the dimension of game complexity (GC for short), we utilize the key elements of game complexity to create the items based on previous research [30,69]. A game is considered complex if it has an excessive number of game elements and element types (items GC0001 and GC0002), intricate relationships between elements (item GC0003), an overwhelming amount of information (item GC0004), or if the information provided is too vague (item GC0005). Similarly, if the game rules are complex (item GC0006) or difficult to understand (item GC0007), or if the goals of the game are unclear (item GC0008) or too numerous (item GC0009), the game may also appear complex. Finally, we include an item that describes the overall perception of game complexity (item GC0010).

For the dimension of game completion difficulty (GCD for short), we created the items that describe players' general feelings of difficulty about the game's challenges (item GCD0001), understanding (item GCD0002), mastering (item GCD0003), and completion (item GCD0004). We also created time-related items: GCD0004 describes the game as highly time-demanding, and GCD0005 describes a game that requires the player to spend much time to complete. GCD0006 and GCD0007 are about the difficulty of winning or achieving the game's preset goal. We revised the items from the NASA-TL scale [51] to create

Table 2
Initial items of the dimensions of game complexity, game completion difficulty, and game-playing difficulty.

Item NO.	Dimensions	Key elements	Initial items
GC0001	Game	Elements	I think the number of game elements is large.
GC0002	Complexity	Elements	I think the types of game elements are too many.
GC0003		Elements	I think relationships among game elements are complex.
GC0004		Information	I think the information provided by this game is too much.
GC0005		Information	I think the information on this game is very vague.
GC0006		Rules	I think the rules of this game are complex.
GC0007		Rules	I think the rules of this game are hard to understand.
GC0008		Goals	I think the goal of this game is unclear.
GC0009		Goals	I think this game contains too many goals.
GC0010		Complexity	This game is very complex.
GCD0001	Game	Challenge	This is a very challenging game.
GCD0002	Completion	Understanding	This game is very difficult to understand.
GCD0003	Difficulty	Master	This game is very difficult to master.
GCD0004		Completion	This game is very difficult to complete.
GCD0005		Time	Completing this game in a demanding time is impossible.
GCD0006		Time	Completing this game needs to take too much time.
GCD0007		Success	This game looks impossible to win.
GCD0008		Success	The goal of this game is unachievable.
GCD0009		Demands	This game is highly mentally demanding.
GCD0010		Demands	This game is highly physically demanding.
GPD0001	Game-playing	Perception	I had to observe very carefully when playing this game.
GPD0002	Difficulty	Perception	I had to identify different things carefully in this game playing.
GPD0003		Memory	l had to memorize a lot of different things when playing this game.
GPD0004		Memory	I had to manage a lot of things at the same time when playing this game.
GPD0005		Thinking	l had to measure each decision carefully I made in this game.
GPD0006		Thinking	l had to think carefully about how to win this game.
GPD0007		Thinking	Thinking fast was an important part of playing this game.
GPD0008		Thinking	Managing time was an important part of playing this game.
GPD0009		Action	I had to act quickly when playing this game.
GPD0010		Action	Playing this game demanded precision in my actions.

Table 3
Initial items of the dimensions of player competence, player pressure, and player effort.

Item NO.	Dimensions	Key elements	Initial items
PC0001	Player	Competent	I feel competent in this game.
PC0002	Competence	Competent	I have no confidence in this game.
PC0003		Competent	I feel very capable and effective in this game.
PC0004		Skill	My game ability is well-matched with this game's challenges.
PC0005		Skill	I think I am pretty good at this game.
PC0006		Skill	I am pretty skilled in this game.
PC0007		Performance	I am satisfied with my performance at this game.
PC0008		Performance	I think I was very successful in accomplishing this game.
PC0009		Comparison	I think I did pretty well in this game, compared to other players.
PC0010		Comparison	I am better than average in this game.
PP0001	Player	Nervousness	I felt very tense while playing this game.
PP0002	Pressure	Nervousness	The actions demanded in this game made me nervous.
PP0003		Fatigue	I feel very exhausted after playing this game.
PP0004		Fatigue	This game made me feel too tired to continue.
PP0005		Anxiety	This game made me anxious about the time.
PP0006		Anxiety	The possible loss of this game made me anxious.
PP0007		Pressure	The stress of this game was beyond my scope.
PP0008		Pressure	I felt very pressured while playing this game.
PP0009		Frustration	I felt very frustrated while playing this game.
PP0010		Frustration	Playing this game made me very discouraged.
PE0001	Player	Attention	I was very focused on playing this game.
PE0002	Effort	Attention	This game kept me on my toes.
PE0003		Effort	Playing this game required me to put great effort.
PE0004		Effort	I put much effort into this game.
PE0005		Energy	I invested much energy into this game.
PE0006		Energy	Playing this game required me to spend a lot of energy.
PE0007		Attempt	I tried very hard to make my actions correct in this game.
PE0008		Attempt	I tried very hard on this game.
PE0009		Attempt	To win this game, I performed my best.
PE0010		Attempt	I tried to give my best performance in this game.

the items GCD0009 and GCD0010 that describe the mental and physical demands of the game.

For the dimension of game-playing difficulty (GPD for short), most items were adapted from the CORGIS by Denisova et al. [8]. We further determined four key difficulty-related elements of game-playing: perception, memory, thinking, and action, based on the human cognitive and behavior process [70]. Regarding perception, the difficulty is in observing (item GPD0001) and identifying things (item GPD0002) in the game. Regarding memory, players may face challenges when they are required to remember many things (item GPD0003). Managing things in the game simultaneously relies on the ability of working memory, and it is also difficult to do (item GPD0004). In addition, decision-making (item GPD0005), thinking carefully (item GPD0006), thinking speed (item GPD0007), and time-thinking (item GPD0008) in

Table 4
Fleiss's Kappa of the six dimensions and hit ratios of 60 initial items (N = 15). The R mark after the item number means this item is removed based on the Kappa analysis results.

Dimensions	Fleiss's Kappa	Items	Hit ratios	Items	Hit ratios
Game Complexity	0.654	GC0001 (R)	0.8	GC0006	1
		GC0002	1	GC0007 (R)	0.8
		GC0003	1	GC0008 (R)	0.87
		GC0004	1	GC0009	0.93
		GC0005 (R)	0.73	GC0010 (R)	0.87
Game Completion	0.454	GCD0001 (R)	0.27	GCD0006	0.8
		GCD0002 (R)	0	GCD0007	0.87
		GCD0003 (R)	0.13	GCD0008	0.87
		GCD0004	0.87	GCD0009 (R)	0.07
		GCD0005 (R)	0.67	GCD0010 (R)	0
Game-playing	0.398	GPD0001	1	GPD0006 (R)	0.67
		GPD0002	0.93	GPD0007	0.8
		GPD0003 (R)	0.67	GPD0008 (R)	0.73
		GPD0004 (R)	0.67	GPD0009	0.87
		GPD0005 (R)	0.53	GPD0010	1
Player Competence	0.703	PC0001	1	PC0006	1
		PC0002 (R)	0.4	PC0007	1
		PC0003	1	PC0008 (R)	0.73
		PC0004 (R)	0.93	PC0009	1
		PC0005 (R)	0.93	PC0010	1
Player Pressure	0.580	PP0001	1	PP0006	1
		PP0002 (R)	0.87	PP0007	0.93
		PP0003 (R)	0.8	PP0008	1
		PP0004 (R)	0.87	PP0009 (R)	0.87
		PP0005	0.93	PP0010	0.93
Player Effort	0.626	PE0001 (R)	0.67	PE0006 (R)	0.6
		PE0002 (R)	0.13	PE0007 (R)	0.6
		PE0003	0.87	PE0008	1
		PE0004	1	PE0009	0.87
		PE0005	1	PE0010	0.87

the game are all related to the difficulty of thinking. Regarding action, it is challenging for players to act quickly (item GPD0009) and precisely (item GPD0010) in the game.

For the dimension of player competence (PC for short), items were partially adapted from the PENS scale [60], the IMI scale [10], and by referring to the self-efficacy theory [50]. Players have competence feelings in the game (items PC0001, PC0002, and PC0003), based on assessing: (1) their skills in this game (PC0004, PC0005, and PC0006), (2) their performance (items PC0007 and PC0008), and (3) their performance compared to other players (items PC0009 and PC0010).

For the dimensions of player pressure (PP for short) and player effort (PE for short), items were partially adapted from the NASA-TL scale [51], the IMI scale [10], and the CORGIS scale [8]. Referring to the emotion category [71], we classified the difficulty-related negative emotions into nervousness (items PP0001 and PP0002), fatigue (items PP0003 and PP0004), anxiety (items PP0005 and PP0006), pressure (items PP0007 and PP0008), and frustration (items PP0009 and PP0010) to create related items. The items in the player effort dimension were created or adapted by considering four aspects: when playing, whether the player (1) pays high attention (items PE0001 and PE0002), (2) invests much effort (items PE0003 and PE0004), (3) puts much energy (items PE0005 and PE0006), and (4) tries their best (items PE0007, PE0008, PE0009, and PE0010).

3.2. Stage ii: Scale development

We conducted an investigation to verify the construct validity and exclude the ambiguous items. We first developed the 60 initial items in the Chinese and Japanese language versions based on the English version. The three versions were developed to improve our SGD measure's universal applicability; professional workers handled the translation between the three languages. Then, we recruited experienced game players for each version to conduct the card-sorting investigation [68]. The card-sorting method asks different judges to sort items into preset categories, which can be used to determine the ambiguous items by the level of agreement.

All the participants (13 males and 2 females, 5 for each language version) were recruited online and finished the investigation through the Google Form website.[1] Our participants were between 19 and 59 years old (M = 28.27, SD = 9.33), while their game experience ranged from 8 to 30 years (M = 16.33, SD = 5.51). All our participants rated their game skills as either skillful or master level. Their favorite game genres were Role-playing games (RPG) (13 of 15) and Multiplayer online battle arena (MOBA) games (10 of 15). The nationalities of our participants were the United States of America (5 of 15), Japanese (5 of 15), and Chinese (5 of 15); they used English, Chinese, or Japanese as their first language.

The participants were asked to classify the 60 initial items into six dimensions: (a) game complexity, (b) game completion difficulty, (c) game-playing difficulty, (d) player competence, (e) player pressure, and (f) player effort. The items were presented in random order. Each item could be assigned to only one of the six dimensions. An introduction to SGD and definitions of the six dimensions were provided before the formal investigation. A detailed example was also provided to illustrate the sorting procedure (see Appendix A).

We calculated the Fleiss's Kappa [72] of each dimension by SPSSAU,[2] which is the online access version of SPSS. Fleiss's Kappa can be used to replace Cohen's Kappa for more than two judges. We also calculated the hit ratios of the 60 initial items; this is based on how many participants select the dimension that matches our preset dimension [68]. Table 4 shows the Kappa values of the six dimensions and hit ratios of the 60 items.

We carefully reviewed the classification results of the items that had low hit rates (lower than 0.9) and removed those items that confused participants; see Table 4. For the dimension of game complexity, we deleted the items GC0001, GC0005, GC0007, GC0008, and GC0010.

[1] https://docs.google.com/forms/
[2] https://spssau.com/

Table 5
First version of SGDS (Total number of items = 33) and Fleiss's Kappa of the six dimensions ($N = 15$).

Old item NO.	New item NO.	Dimensions	Fleiss's Kappa	Items
GC0002	GC1001	Game Complexity	0.946	I think the types of game elements are too many.
GC0003	GC1002			I think relationships among game elements are complex.
GC0004	GC1003			I think the information provided by this game is too much.
GC0006	GC1004			I think the rules of this game are complex.
GC0009	GC1005			I think this game contains too many goals.
GCD0002	GC1006			This game is very difficult to understand.
GCD0004	GCD1001	Game Completion Difficulty	0.822	This game is very difficult to complete.
GCD0006	GCD1002			Completing this game needs to take too much time.
GCD0007	GCD1003			This game looks impossible to win.
GCD0008	GCD1004			The goal of this game is unachievable.
GPD0001	GPD1001	Game-playing Difficulty	0.787	I had to observe very carefully when playing this game.
GPD0002	GPD1002			I had to identify different things carefully in this game playing.
GPD0007	GPD1003			Thinking fast was an important part of playing this game.
GPD0009	GPD1004			I had to act quickly when playing this game.
GPD0010	GPD1005			Playing this game demanded precision in my actions.
PC0001	PC1001	Player Competence	0.904	I feel competent in this game.
PC0003	PC1002			I feel very capable and effective in this game.
PC0006	PC1003			I am pretty skilled in this game.
PC0007	PC1004			I am satisfied with my performance at this game.
PC0009	PC1005			I think I did pretty well in this game, compared to other players.
PC0010	PC1006			I am better than average in this game.
PP0001	PP1001	Player Pressure	0.906	I felt very tense while playing this game.
PP0005	PP1002			This game made me anxious about the time.
PP0006	PP1003			The possible loss of this game made me anxious.
PP0007	PP1004			The stress of this game was beyond my scope.
PP0008	PP1005			I felt very pressured while playing this game.
PP0010	PP1006			Playing this game made me very discouraged.
PE0003	PE1001	Player Effort	0.882	Playing this game required me to put great effort.
PE0004	PE1002			I put much effort into this game.
PE0005	PE1003			I invested much energy into this game.
PE0008	PE1004			I tried very hard on this game.
PE0009	PE1005			To win this game, I performed my best.
PE0010	PE1006			I tried to give my best performance in this game.

For the dimension of game completion difficulty, we removed items GCD0001, GCD0002, GCD0003, GCD0005, GCD0009, and GCD0010. Interestingly, all the participants classified GCD0002 into the game complexity dimension, so we moved this item into that dimension. Items GPD0003, GPD0004, GPD0005, GPD0006, and GPD0008 in the game-playing difficulty dimension were also removed. For the dimension of player competence, we deleted items PC0002, PC0004, PC0005, and PC0008. Items PP0002, PP0003, PP0004, and PP0009 in the player pressure dimension and items PE0001, PE0002, PE0006, and PE0007 in the player effort dimension were also removed for the low level of agreement between judges. We recalculated Fleiss's Kappa of the six dimensions and listed the first version of our SGD scale in Table 5. After removing these items, all the dimensions have Kappa values larger than 0.8. The game-playing difficulty dimension, with the value of 0.787, was considered close to 0.8 and was acceptable. This means that all the dimensions achieve high consistency among the judges. Therefore, we believe the instrument reached a good quality for testing in the next stage.

3.3. Stage iii: Scale testing

This stage aimed to test the reliability and validity of our scale through an online survey. Because our SGD scale should be measured based on player experience with a specific game, the testing in this stage could only focus on the players' SGD in one game. Therefore, to test our developed scale effectively, we selected the testing game by these criteria: (1) this game is popular with many players, (2) to avoid extreme test results, this game should not be too difficult (or too complex) or too easy, and (3) this game demands focus on the action, thinking, and playing time.

Referring to the research that introduced typical game genres [73–75], we first identified 16 main game genres; see Table 6. We excluded games without explicit success or failure goals (simulation games and fiction games), without stable difficulty (gambling games), or games that were too complex (RPG, multiplayer online games). The games that focus solely on the gameplay of action or thinking were also excluded (shooter games, action games, music games, sports games, racing games, platformer games, strategy games, puzzle games, and board and card games). Therefore, we confirmed the genres of real-time strategy (RTS) games and casual games. However, RTS games are famous for their difficulty [76], and some types of casual games are too easy [77]. Therefore, we settled on tower defense games, which are a kind of casual game but still challenging enough [78]. Additionally, tower defense games demand players to both act quickly and think about the locations of defense units and to consider the remaining time. It should be clarified that our selection of the specific game type for scale testing does not mean the scale cannot be used in the measurements of other games. We discuss how the developed scale can be applied in other games in Section 4.1.1.

We searched the tower defense games on the Steam store [79], which is one of the most popular game distribution platforms on personal computers (PC) [80]. All the pages of this online game store were accessed in April 2024. After searching from the top 10 games that appeared in the "Top sellers" or "Top rated" lists of the tower defense genre, we finally chose the *Plants vs. Zombies* game [81]. *Plants vs. Zombies* is a popular tower defense game and has sold over 30 million copies. The gameplay of this game requires players to collect sunlight resources in order to set plants to fight against zombies, and it involves thinking, action, and time pressure; see Fig. 4. Therefore, this game perfectly satisfies our criteria.

To ensure our participants have played the game, we set five questions (see Appendix A.2) before the formal survey, which they must correctly answer to prove they have become familiar with the game. Only those participants who correctly answered all five questions could attend the formal survey. The formal survey contained two parts: the first part collected basic information about participants, and the second

Table 6
List of 16 typical game genres.

Game genres	Brief descriptions
Shooter Games	Games of shooting by weapons, usually firearms.
Action Games	Games that emphasize fast actions and reactions.
Role-playing Games (RPG)	Games that require players to take on an in-game role.
Music Games	Games that require players to act with music.
Sports Games	Games that simulate one or more real-life sports.
Multiplayer Online Games	Games that require multiple players to cooperate and compete online.
Racing Games	Games featuring the driving of vehicles for competition.
Platformer Games	Games that require players to jump between platforms.
Strategy Games	Games that require players to make decisions strategically.
Real-time Strategy Games (RTS)	Games that require players to make decisions strategically in real-time.
Simulation Games	Games of simulating real-world activities.
Board and Card Games	Games that use cards and pieces (e.g., tiles) to play.
Gambling Games	Games that require players to wager.
Puzzle Games	Games that require players to solving puzzles.
Fiction Games	Games that require players to progress story.
Casual Games	Games with simple rules, shorter sessions, and that require less learned skill.

Fig. 4. Screenshots of the *Plants vs. Zombies* game. The gameplay of this game is to collect sunlight resources to set various plants to defend against zombies, which imposes time pressure and demands players to think clearly and act quickly.

part was a 35-item Likert questionnaire (using the first version of the 33-item SGD scale plus two polygraph questions). The questions in this questionnaire were shown in random order, and responses ranged from 1 (strongly disagree) to 7 (strongly agree). The survey was in three language versions, namely, English, Chinese, and Japanese; professional workers handled the translation between the various languages.

All the participants were recruited online through Amazon Mechanical Turk,[3] CloudWorks,[4] and Wenjuanxing.[5] 565 responses were collected, but only 342 participants correctly answered the five preset questions and completed the formal survey. Sixteen participants were removed for answering the polygraph questions incorrectly or not providing adequate demographic information. Finally, the data from 326 participants (175 males and 151 females) were used in the analysis. Our participants were aged from 18 to 60 ($M = 26.23$, $SD = 7.29$); their nationalities were the United States of America (134 of 326), China (111 of 326), and Japan (81 of 326); they used English, Chinese, or Japanese as their first language.

Table 7
Internal consistency (Cronbach's α) of SGDS ($N = 326$).

Dimensions	Cronbach's α
Game Complexity	0.912
Game Completion Difficulty	0.852
Game-playing Difficulty	0.830
Player Competence	0.896
Player Pressure	0.902
Player Effort	0.806

The game experience of our participants ranged from 1 to 45 years ($M = 11.25$, $SD = 7.94$). For playing frequency and game skills, most of our participants had played games within a few days (203 of 326) or a few days ago (59 of 326), and rated their game skills as ordinary (114 of 326) or skillful (99 of 326). All the participants have played the genre of casual games before, and the other game genres they most preferred were role-playing games (RPG) (169 of 326) and action games (186 of 326). In addition, most of our participants had a game time of *Plants vs. Zombies* of 1–5 h (137 of 326) or within 1 h (134 of 326), and rated their game skills in this game as ordinary (127 of 326) or skillful (85 of 326).

[3] https://www.mturk.com/
[4] https://crowdworks.jp/
[5] https://www.wjx.cn/

Table 8
The standard factor loading of all the items (Total number of items = 33) in of our first version SGD scale through a CFA (N = 326).

NO.	Dimensions (factors)	Items	Std. factor loading
GC1001	Game	I think the types of game elements are too many.	0.802
GC1002	Complexity	I think relationships among game elements are complex.	0.797
GC1003		I think the information provided by this game is too much.	0.818
GC1004		I think the rules of this game are complex.	0.791
GC1005		I think this game contains too many goals.	0.777
GC1006		This game is very difficult to understand.	0.792
GCD1001	Game	This game is very difficult to complete.	0.796
GCD1002	Completion	Completing this game needs to take too much time.	0.599
GCD1003	Difficulty	This game looks impossible to win.	0.826
GCD1004		The goal of this game is unachievable.	0.879
GPD1001	Game-playing	I had to observe very carefully when playing this game.	0.775
GPD1002	Difficulty	I had to identify different things carefully in this game playing.	0.733
GPD1003		Thinking fast was an important part of playing this game.	0.611
GPD1004		I had to act quickly when playing this game.	0.632
GPD1005		Playing this game demanded precision in my actions.	0.748
PC1001	Player	I feel competent in this game.	0.819
PC1002	Competence	I feel very capable and effective in this game.	0.737
PC1003		I am pretty skilled in this game.	0.798
PC1004		I am satisfied with my performance at this game.	0.700
PC1005		I think I did pretty well in this game, compared to other players.	0.829
PC1006		I am better than average in this game.	0.742
PP1001	Player	I felt very tense while playing this game.	0.831
PP1002	Pressure	This game made me anxious about the time.	0.668
PP1003		The possible loss of this game made me anxious.	0.719
PP1004		The stress of this game was beyond my scope.	0.839
PP1005		I felt very pressured while playing this game.	0.822
PP1006		Playing this game made me very discouraged.	0.789
PE1001	Player Effort	Playing this game required me to put great effort.	0.753
PE1002		I put much effort into this game.	0.820
PE1003		I invested much energy into this game.	0.768
PE1004		I tried very hard on this game.	0.446
PE1005		To win this game, I performed my best.	0.552
PE1006		I tried to give my best performance in this game.	0.422

Table 9
Second CFA (N = 326) after removing some items of our first version SGDS (Total number of items = 25). Results of the standard factor loading.

NO.	Dimensions	Std. factor loading	NO.	Dimensions	Std. factor loading
GC1002	Game	0.791	PC1001	Player	0.818
GC1003	Complexity	0.796	PC1002	Competence	0.736
GC1004		0.796	PC1003		0.798
GC1006		0.828	PC1004		0.698
GCD1001	Game	0.791	PC1005		0.830
GCD1003	Completion	0.838	PC1006		0.743
GCD1004	Difficulty	0.889	PP1001	Player	0.836
GPD1001	Game-playing	0.775	PP1004	Pressure	0.865
GPD1002	Difficulty	0.732	PP1005		0.805
GPD1003		0.611	PP1006		0.782
GPD1004		0.629	PE1001	Player Effort	0.775
GPD1005		0.749	PE1002		0.819
			PE1003		0.790

3.3.1. Reliability testing

We employed Cronbach's α [82] to measure the reliability, and the analysis was conducted by SPSSAU. The Cronbach's α was calculated for the six dimensions using all the collected data. The results show the six dimensions all have Cronbach's α larger than 0.8, see Table 7. Therefore, the internal consistency of each dimension is acceptable.

3.3.2. Validity testing

A confirmatory factor analysis (CFA) was done on the survey data using the 6-factor structure based on the six dimensions for validity testing. CFA is commonly used to analyze the efficacy of measuring models and test the scale's validity [83]. The standard factor loading by CFA is shown in Table 8. Based on the literature [83], the estimated value of factor loading should be larger than 0.6, or it means the correlation between this item and the factor is not strong enough.

We carefully checked all the items with a value less than 0.8 and removed items GC1001, GC1005, GCD1002, PP1002, PP1003, PE1004, PE1005, and PE1006 to improve the quality of the measurement model. Items GPD1003 (factor loading = 0.611) and GPD1004 (factor loading = 0.623) were retained even with low loading values because this dimension measures different aspects of game-playing, and we thought that values were still acceptable.

After removing these items, a second CFA was conducted; see Tables 9 and 10. According to the literature [84], the chi-squared statistic and other model fit indices (χ^2 is significant and $\chi^2/df < 3$; CFI > 0.9, TLI > 0.9, RNI > 0.9, SRMR < 0.1, RMSEA < 0.08) should be used to confirm whether the model fit is acceptable. As the result shows, the chi-squared statistic ($\chi^2 = 643.34$, df = 260, $p < 0.001$; $\chi^2/df = 2.474$) and the model fit indices (CFI = 0.930, TLI = 0.919, RNI = 0.921, SRMR = 0.065, RMSEA = 0.067) demonstrate that the CFA model is acceptable. We found that all the items had a value of factor loading that was larger than 0.6 and showed good correlation relationships between the items and dimensions.

Based on the result of the second CFA, we further analyzed the validity of this scale by calculating the Composite Reliability (CR) and Average Variance Extracted(AVE). According to the literature [84], the

Table 10
Second CFA ($N = 326$) after removing some items of our first version SGDS (Total number of items = 25). Results of the model fit index. Note: ***Denotes significant at the 0.001 levels (2-tailed).

χ^2	df	p	CFI (> 0.9)	TLI (> 0.9)	RNI (> 0.9)	SRMR (< 0.1)	RMSEA (< 0.08)
643.34***	260	< 0.001	0.930	0.919	0.921	0.065	0.067

Table 11
Composite Reliability (CR) and Average Variance Extracted (AVE) of the six dimensions ($N = 326$).

Dimensions	Average Variance Extracted (AVE)	Composite Reliability (CR)
Game Complexity	0.645	0.879
Game Completion Difficulty	0.706	0.878
Game-playing Difficulty	0.493	0.828
Player Competence	0.596	0.898
Player Pressure	0.676	0.893
Player Effort	0.632	0.838

Table 12
Pearson correlation coefficient r between each pair of dimensions and each dimension's square root of AVE ($N = 326$).

	GC	GCD	GPD	PC	PP	PE	Square root of AVE
GC	1						0.803
GCD	0.796	1					0.840
GPD	0.508	0.405	1				0.702
PC	0.174	−0.006	0.360	1			0.772
PP	0.802	0.785	0.481	0.089	1		0.822
PE	0.672	0.559	0.657	0.368	0.657	1	0.795

Table 13
Invariance testing (Total $N = 326$: N of English = 134, N of Chinese = 111, N of Japanese = 81).

	Tucker's $\phi 1$	Tucker's $\phi 2$	correlation 1	correlation 2
English	0.96	0.92	0.97	0.96
Chinese	0.96	0.95	0.96	0.98
Japanese	0.99	0.97	0.91	0.90

convergent validity of the measurement will be good when AVE values are larger than 0.5, and CR values are larger than 0.7. The result in Table 11 shows that all the values of AVE and CR fit the standard except the AVE of game-playing difficulty, which had a value of 0.493. The reason might be that we retained the items GPD03 and GPD04. However, the value was very close to 0.5, which we believe was still acceptable.

We also conducted a Pearson correlation analysis to calculate the coefficient r between each pair of dimensions, see Table 12. According to the literature [84], if the square root of AVE is larger than the dimension's r with other dimensions, the discriminant validity of that dimension is good. The result showed that dimensions of GC (0.803 > 0.802), GCD (0.840 > 0.796), GPD (0.702 > 0.657), PC (0.772 > 0.366), PP (0.822 > 0.802), and PE (0.795 > 0.672) had their AVE's square root values larger than the related r values with other dimensions. Therefore, the discriminant validity of SGDS with these six dimensions could also be rated as good.

3.3.3. Invariance testing

Because we conducted the survey in three language versions, it is necessary to test the invariance between different versions. Based on research by Fischer and Karl [85], we conducted a multi-group CFA in R and calculated the values of Tucker's ϕ and the correlation coefficient between the three language groups. The two values that are closer to 1 represent that these groups are more similar to each other. The result showed a minor but acceptable invariance between these three language groups, see Table 13.

4. Discussion

In this paper, we develop a new scale to measure players' subjective game difficulty (SGD). This scale has been verified for its reliability and validity. This section summarizes the usability of this scale and compares it with other SGD measurements. In addition, this section discusses the implications of this study for game design research and provides an introduction to the usage of this instrument.

4.1. Subjective game difficulty scale (SGDS)

4.1.1. Usability of SGDS

This work develops a scale with six subscales to measure the multidimensional structure of SGD, i.e., subjective game difficulty scale (SGDS). According to the results in Section 3.3, the consistency within each subscale is high (in reliability); the items are highly correlated within each subscale (convergent validity), and the items are also highly relevant to their respective subscale (structural validity). Therefore, the developed scale has been proven to be a qualified measurement instrument.

In the development process of SGDS, we deleted some ambiguous and ineffective items. One possible question is whether the remaining items in each subscale can fully represent the expected dimensions, i.e., whether the content validity is still guaranteed. We reviewed the corresponding key elements of the both retained and removed items in each subscale (see Tables 2 and 3). For the dimension of game complexity (GC), the retained items reflect complexity from the aspects of elements, information, and rules, while the items describing goals and complexity are excluded. For the game completion difficulty (GCD) dimension, the retained items are about completion, success, and goals; the understanding-related item is moved to the GC dimension, and items of challenge, mastery, time, and demands are excluded. For the game-playing difficulty (GPD) dimension, items of perception, thinking, and action are retained, while memory-related items are excluded. For the player competence (PC) dimension, all items of the four elements originally proposed are retained. For the player pressure dimension (PP), the items describing the feelings of nervousness, pressure, and frustration are retained, but the items representing anxiety and fatigue are excluded. Finally, for the player effort (PE) dimension, the items of effort and energy are retained, but the items of attention and attempt are excluded.

We found that most abandoned items were related to more than one dimension and thus tended to be ambiguous. For example, the element of goals is contained in the GCD dimension, so the goal-related items in the GC are removed (GC0008 and GC0009); the items describing time in the GCD dimension are excluded (GCD0005 and GCD0006) because they partially overlap with the action element of the GPD dimension. Similarly, the elements of attention and attempt in the PE dimension partially overlap with the performance in the PC dimension. For the GPD dimension, memory-related items are excluded (GPD0003 and GPD0004), maybe because such demands are not obvious in most games. For the PP dimension, fatigue may partially overlap with the energy element in the PE dimension. As for the items of anxiety in the PP dimension (PP0005 and PP0006), they may not be common in most player groups [34], although it is a common negative emotion. Therefore, the items deleted were all ambiguous or ineffective. By excluding these items, the subscales of SGDS are well separated from each other and gain better content validity to represent the six SGD dimensions.

For the application scope of SGDS, SGD is suitable for the games that involve clear tasks, because game difficulty comes from the interaction between players and game tasks. Our survey results in Section 3.3 are applicable to most genres of games because the selected tower defense game involves general elements of game tasks (i.e., thinking, action, and time pressure). In more detail, SGDS can be applied well in most types of games, except for simulation games, fiction games, and gambling games (see Table 6). The reason is that the gameplay of these games is usually not based on tasks demanding player skills. For instance, as a kind of simulation game, sandbox games allow players to move and construct freely in the game and thus, failure is not possible [86]. Fiction games may have multiple endings, and failure or wrong decisions cannot be strictly defined, so task-based difficulty cannot be measured [87]. Gambling games are also an exception. The tasks of gambling games are usually simple, and the results of the games mostly depend on probability (e.g., the Blackjack game [88]). Therefore, the correlation between task performance and player skills is not strong in some gambling games, which compromises the SGD measurement.

SGDS can also be applied in different stages of measurement. In the first stage of scale development (Section 3.1), we mentioned that the dimensions of GC and GCD can be measured at any stage of the game, and the other four dimensions can be measured during and after the game. This makes SGDS more flexible as it can be applied to measure players' SGDs before, during, and after the game. Therefore, SGDS has a large scope of application. We introduce in detail how to use SGDS for measurement at different stages in Section 4.3.

In summary, the developed SGDS has good usability. The scale has a large scope of application. The subscales of SGDS have good independence and can conceptually represent all the dimensions of SGD well.

4.1.2. SGDS compared to other SGD measuring methods

As mentioned in Section 2.3, there are currently three main SGD-measuring methods: simple self-report, structured self-report, and physiological measurement. Our developed scale is a new instrument that belongs to the structured self-report method. We compared our scale with current methods and instruments to determine how it improves current SGD measurements.

Compared to the simple self-report method, our instrument supports a more detailed and comprehensive measurement of SGD. More specifically, the simple self-report method asks the player to rate whether the game is difficult, usually with scores [21,28,36]. However, as we summarized, the difficulty of the game has different aspects, but this method merely measures each player's overall perception of the game's difficulty. Instead, the developed SGDS uses six dimensions, namely, game complexity, game completion difficulty, game-playing difficulty, player competence, player pressure, and player effort, to measure the SGD of players. Through structured questions in these dimensions, this instrument can help players evaluate the different aspects of game difficulty comprehensively and in more detail.

Physiological measurements have gained increasing interest in recent years [89]. This type of method is more objective and non-intrusive and can be used in real-time. However, this method provides information about players' physiological state, such as emotion and pressure, rather than their thoughts and views [23,58,63]. Therefore, although the state of effort and pressure can be measured more directly, other dimensions of SGD still cannot be determined. We believe the biodata may contain more potential information that can reveal SGD, even if this relationship is currently unclear. Based on advanced AI techniques, this relationship can be clarified through effective measurement and modeling [64–66]. Therefore, by providing a more precise measurement, the developed scale supports the establishment of the relationship between the player's physiological information and SGD.

As a new instrument of the structured self-report method, we compared SGDS with the existing scale, CORGIS [8]. As introduced in Section 2.3, CORGIS is for measuring the players' perceived challenges, providing items from four dimensions that reflect different aspects of the task demands; some items measuring perceived effort are involved. We suggest that these items can be used in measuring the dimensions of game-play difficulty and player effort, but they may miss measuring the four other dimensions. Also, a recent study indicated these challenge types in CORGIS were difficult to distinguish [66]. In comparison, our scale is more comprehensive for measuring the six dimensions of SGD. Furthermore, the CORGIS scale also tried to measure emotional challenge, which is related to narrative tension, identification of game characters, and ambiguities [8,90]. We agree that some scenarios set in games can challenge players emotionally (e.g., moral dilemma). However, we think this experience relates little to game difficulty: there is no clear task that needs to be solved, and no specific demands on player skills. The items like "The things that happened in the game made me sad" and "Playing the game was stimulating" in CORGIS measure emotional experience rather than SGD. Therefore, CORGIS is more adaptable in the measurement of fiction games than SGDS because it can measure emotional experience. However, SGDS can be used to measure player SGD before, during, and after the game, and its application scope is broader than CORGIS.

In summary, the new scale SGDS has a moderate number of items (N=25) with good usability and versatility. It significantly improves the existing methods and is expected to measure players' SGD in a more comprehensive, precise and structured manner.

4.2. Implications for game design research

First, the six dimensions of SGD verified in this study indicate that the players' subjective perceptions of difficulty are complex and influenced by multiple factors. However, OGD is related more to player performance, which may lead to a natural mismatch between SGD and OGD. In other words, the structural differences between SGD and OGD lead to this mismatch. This view is also supported by recent studies, e.g., Guo et al. [26] found that OGD and SGD only partially match each other. We believe that this mismatch is not negative, but implies that rich SGD-related experiences may be triggered even by the same OGD. By identifying this mismatch, designers can adjust OGD to create more varied SGD possibilities to optimize player experience. For example, failures are common and necessary in roguelike games where players need to fail and retry continuously [91,92]. Playing this game genre, players enjoy the different journeys and processes of becoming more skilled, even with the same failure (OGD) in the game.

Second, the six dimensions of SGD may explain different sources of player experience. For example, defeating an enemy that looks overwhelming (game completion difficulty) greatly enhances the player's sense of achievement, even though the process is not as hard as imagined (game-playing difficulty) [1]. In addition, research shows that players tend to be positive about their experience even in high OGD conditions [38]. We believe that players can protect their positive experience by selecting different SGD dimensions for self-explanation. For example, when facing too many failures (player competence), players can attribute their poor performance to the task being too complicated (game complexity) and impossible to complete (game completion difficulty). In short, these six dimensions provide novel perspectives for understanding the player experience.

Third, game difficulty design can also benefit from using our scale. Effective evaluation of player SGD is required for design validation. Compared to other measuring methods, SGDS is a more comprehensive tool that can be applied in design evaluation. Designers can use the scale to validate whether their design creates the expected subjective difficulty experience. Additionally, the proposed six dimensions can also be used to guide SGD design. For example, the serious game *My Cotton Picking Life* [93] demands players to pick cotton by repeating click operations constantly for 6 h, and such a torturous process strengthens persuasive effects (i.e., affects players' attitudes to child slave labor) [94]. This example presents how to create a high level of SGD by focusing solely on the player effort dimension. Finally, awareness of the SGD scores of other players in the game is also an interesting design idea. Because the scores provide a baseline, players may feel proud if they succeed (or less frustrated if they fail) in a high SGD game level.

Finally, the SGDS and theoretical findings of this study also support the design of the dynamic difficulty adjustment (DDA) mechanism. According to Guo et al. [16], DDA is for automatic difficulty control in interaction by the real-time evaluation of OGD and SGD. Therefore, designers can use the SGDS to evaluate the real-time SGD of players to support difficulty adjustment during the game. Moreover, the skill-challenge balance proposed by flow theory is more related to SGD [6, 95]. Some designers have noticed this point and adopted SGD in their DDA design [20,23,28]. Furthermore, the partial matching relationship between OGD and SGD allows designers to adjust the OGD finely with DDA while keeping SGD stable [26]. Results of some DDA research supported this view [96,97]. However, such sophisticated design relies on effective measurements. Therefore, the developed scale supports DDA design and evaluation by providing a more comprehensive SGD measurement.

4.3. How to use SGDS

We provide the final version of the SGD scale in Appendix B (English version in B.1, Chinese version in B.2, and Japanese version in B.3). This scale is a 25-item Likert questionnaire, and we suggest using the range 1 (strongly disagree) to 7 (strongly agree) for measurement. This scale has six subscales; the subscale of player competence is negatively related to the level of SGD, while the other subscales are positively related to it (i.e., the subscales of game complexity, game completion difficulty, game-playing difficulty, player pressure, and player effort). Therefore, the higher the average score of each subscale (the lower the average score of the player competence subscale), the more difficult the player perceives that dimension to be.

SGDS can be used to measure the overall feelings of difficulty by integrating the scores from all the subscales, or to measure the different SGD dimensions by using the corresponding subscales. To use SGDS, it is necessary to first determine the measuring goals and timings. Although it is common to measure SGD after play, our scale also supports the measurement before play and during the game. Usually, the SGD measurement before playing helps identify players' first impressions of the game, while measuring SGD during the game helps designers validate their OGD design. We suggest how to use SGDS in the three stages – before, during, and after the game – as follows.

In the before-playing stage, the players may have only a rough idea after observing the game. Because they have not yet played, players cannot evaluate the interaction process or their state, so they will not have generated reliable related SGD feelings. However, they can still assess the two dimensions of game complexity and game completion difficulty roughly based on their first impressions of the game. One of the values of the SGD measurement in this stage is identifying players' first impressions of the game's difficulty. This impression may affect their initial willingness to play this game and their subsequent play experience [98]. This measurement is useful for the market research of a new game. This is because it is important to know whether the target player group has the expected SGD feelings and is motivated to purchase [99].

SGDS can also be used in player SGD measurement during the game. However, rather than a real-time and non-invasive measurement tool, SGDS needs to be used at intervals during the game (such as game pauses). Measuring the changes of SGD during the game is necessary in some research and game design. For example, designers draw player difficulty curves [16] or model SGD based on players' data [64,89]. Additionally, for games with adaptive difficulty mechanisms (i.e., DDA games), designers need to confirm the SGD change as difficulty adjusts. To measure SGD by SGDS during the game, designers need to determine the timing, frequency, and items used in the measurement. The measurement timing depends on the design needs, usually before and after the game content changes. Secondly, a too high frequency of measurement is not preferable because it would split the continuous game process. Finally, answering too many items may also affect the fluency of players' experiences. Therefore, we suggest selecting parts of subscales and eliminating some items accordingly. For example, to measure the SGD of DDA games, designers may employ a repeated measurement using SGDS during the game. The measurement timing can be set to the point after the system adjusts the game difficulty. The subscales of game complexity and game-playing difficulty can be used. The measurement frequency should be equal to the difficulty adjustment frequency. One possible result is that, due to their skill enhancement, players may notice the increase in game complexity caused by adjustment (the scores on this subscale increase), but still perceive the game-playing difficulty as moderate (the scores on this subscale remain stable).

In the after-playing stage, SGDS can be used to measure the player's feelings of difficulty about their present game experience or previous game experiences (recent is not necessary). In comparison, the CORGIS scale emphasizes that its measured perceived challenge should originate from their recent gameplay [8]. For research, researchers can use SGDS in their experiments as a post-test in game research or measure players' SGD of a game through an online survey using SGDS, based on their requirements. However, since most games are large and have various game levels with different levels of difficulty, it is necessary to clearly specify the SGD evaluation scope when using SGDS. Researchers need to inform players that the SGD evaluation scope is the whole game, a certain game level, a certain period of their playing, etc. In addition, SGDS can be used in comparison research, e.g., the differences in difficulty preferences between player groups. This scale can also be combined with other scales (e.g., PENS [10]) to study the interactive relationships between SGD and other factors [26].

5. Limitations & future work

This study developed and validated a new scale for SGD. Data from 326 participants, using English, Chinese, or Japanese, were collected and included in the validation testing, which is still limited. Future work includes data collection from additional participants responding in these three languages and further validation. Considering the cultural differences among countries, international comparisons on this issue present a useful and informative challenge for future research. Further validation of this instrument is expected to solve the issues of minor invariance between the three language versions of the scale.

As the first step in our research, we validated our scale in one game. To ensure the scale's broader utility, further validation across additional genres and game types is necessary. Surveys on SGDS applied in the measurement of non-causal games are required in the future [8]. As we mentioned in Section 4.1, the scale has limitations in genre-specific applicability. It cannot be used in measuring SGD in simulation games, fiction games, and gambling games. Further research is necessary for measuring SGD in these games in the future. In addition, reliance on subjective questionnaire surveys based on self-reports has inherent limitations. For example, such methods cannot be applied in a real-time or non-invasive manner [28]. Moreover, several psychological biases, such as response style biases, memory and recall biases, and cognitive biases, may influence the results of self-report [100,101]. Finally, future studies should also address the challenge of SGD measurement differences across various game types. Our findings may encourage researchers to further develop SGDS measurement methods from both objective and subjective perspectives, and to apply these methods across different types of games.

Furthermore, how SGDS can be applied in games with the DDA mechanism also remains to be verified. Due to the limitations of the self-report method, measuring changes in SGD in real-time using SGDS is difficult. Determining the suitable timing, frequency, and items for SGD measuring should be included in future DDA games' measurement studies. Moreover, the adjustment by DDA may be subtle and challenging to capture the slight changes of SGD using the self-report method. Therefore, it is necessary to employ qualitative methods (e.g., interviews) to gain deeper insights into how DDA affects SGD and to explore real-time SGD measurement in DDA games.

Validation comparison studies of SGDS, CORGIS, and other metrics are included in future work. The CORGIS measurement includes the type of emotional challenge. Although emotional challenges have been recently studied, we excluded this type of challenge from our scale development. The reason is that the current concept of emotional challenge is not suitable for quantification and measurement in SGD. Further clarification and quantification work are promising to make this concept more operational. However, we believe this concept has the potential to support the future improvement of our SGD scale.

Finally, although the developed SGDS can effectively measure the multi-dimensional structure of SGD, it still requires iteration and improvement to enhance its quality. Therefore, expanding the current version of the scale by adding reliable and validated items is included in our future plans. Furthermore, empirical research that verifies the proposed implications in game design research is encouraged. For example, further study of how the subscale scores of SGDS relate to different aspects of player experience would be informative. We also encourage researchers to apply this scale to support the development of SGD physiological measurements.

6. Conclusion

To measure subjective game difficulty (SGD) effectively and structurally, this work developed and validated the Subjective Game Difficulty Scale (SGDS). This new scale measures the six dimensions of SGD: game complexity, game completion difficulty, game-playing difficulty, player competence, player pressure, and player effort. We adopted the standard three-stage development method, namely, item generation, scale development, and scale testing, to develop this new measurement instrument. An original international survey was conducted for scale testing in the third stage. The results show that our final version of the SGDS with 25 items had good reliability and validity, and that it is valuable for future SGD measurement. We discussed the usability of SGDS and its differences with other measuring methods. We also proposed the implications for game design research and made suggestions on how our scale can be applied. Limitations and future plans were also provided. We expect that this new scale will become an important instrument to support SGD measurement and evaluation for researchers and designers.

CRediT authorship contribution statement

Zhixing Guo: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Xiangshi Ren:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Conceptualization. **Xinxin Ma:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Table 14
The definitions of the six dimensions.

Dimensions	Definitions
Game Complexity	The player's perception of the game's attributes that may cause difficulty. Players can describe these attributes without the game playing.
Game Completion Difficulty	The player's general perception of the game's completion difficulty. Players can describe this difficulty without the game playing.
Game-playing Difficulty	The player's perception of the actual difficulties they faced while playing. These difficulties are caused by the game's demands on players' skill level and are experienced by players during the game playing. This type of difficulty can be described after the game playing.
Player Competence	The player's perception of their performance and competence in playing this game. Playing experience in this game is necessary for players' competence self-evaluation. Therefore, competence is usually evaluated after the game playing.
Player Pressure	The player's perception of stress and other accompanying negative feelings from completing the game. Pressure feeling is experienced during game playing and is usually evaluated after the game playing.
Player Effort	The player's perception of their effort and investment in completing the game. Effort feeling is experienced during game playing and is usually evaluated after the game playing.

Table 15
Questions about *Plants VS. Zombies* before the formal survey.

NO.	Question contents	Types	Options
1	In the game, the resource needed to grow plants is:	Single choice	A Coins, B. Sunlight C. Water, D. Seeds
2	In the game, there is a type of plant that needs to to be planted in night conditions; they are:	Single choice	A. Fruits, B. Flowers C. Peas D. Mushrooms
3	In the game, when zombies encounter plants, they will ? plants.	Single choice	A. Eat, B. Attack C. Dig, D. Pull-out
4	In the game, what is the name of your neighbor?	Single choice	A. Crazy Jack, B. Crazy Dave C. Crazy Steve, D. Crazy Inventor
5	In the game, the factors you need to consider are:	Multiple choice	A. Types of plants B. Locations of plants C. Order of planting D. Types of zombies E. Routes of zombies F. Quantity of resources

Appendix A. Online survey materials

A.1. Materials of the card sorting procedure

Subjective game difficulty refers to the player's perceptions of difficulty with the game. These perceptions arise from the player's overall evaluation of three aspects: the game's attributes that may cause difficulty, the actual difficulties the player faced while playing, and the player's motivational and emotional states when encountering difficulty in play. We provide you with six dimensions related to subjective game difficulty and introduce these dimensions (see Table 14). We provide the 60 initial items in random order for classification into these dimensions.

Please classify each of the following 60 initial items into one of the six provided dimensions. For instance, if you believe the item "This game is..." pertains to the player's perception of "Game Complexity", select this dimension from the six available options.

You can review the introduction to these dimensions when unsure how one item can be classified, or you can select the "Other" option and provide your opinion. The number of items classified into each dimension does not need to be the same.

A.2. Materials of the scale testing

To attend this survey, you must have played the game "*Plants vs. Zombies*" produced by Popcap Games because we need you to complete this survey based on your experience in this game. Therefore, before the formal survey, you must correctly answer five questions about this game to prove you have played it and are familiar with it (see Table 15).

Appendix B. Final version of the subjective game difficulty scale (SGDS)

B.1. English version of the SGDS

The Subjective Game Difficulty scale (Below are listed all 25 items that can be used as needed. To score this instrument, you must reverse score the items for the Player Competence subscale by subtracting each item response from 8 and score other items as usual. The higher the average score of each subscale, the more difficult the player perceives that dimension to be.)

For each of the following statements, please indicate how much you agree with it, using the following scale:

1-Strongly disagree, 2-disagree, 3-Somewhat disagree, 4-Neutral, 5-Somewhat agree, 6-Agree, 7-Strongly agree

Game Complexity

I think relationships among game elements are complex.
I think the information provided by this game is too much.
I think the rules of this game are complex.
This game is very difficult to understand.

Game Completion Difficulty

This game is very difficult to complete.
This game looks impossible to win.
The goal of this game is unachievable.

Game-playing Difficulty

I had to observe very carefully when playing this game.
I had to identify different things carefully in this game playing.
Thinking fast was an important part of playing this game.
I had to act quickly when playing this game.

Playing this game demanded precision in my actions.

Player Competence

I feel competent in this game.

I feel very capable and effective in this game.

I am pretty skilled in this game.

I am satisfied with my performance at this game.

I think I did pretty well in this game, compared to other players.

I am better than average in this game.

Player Pressure

I felt very tense while playing this game.

The stress of this game was beyond my scope.

I felt very pressured while playing this game.

Playing this game made me very discouraged.

Player Effort

Playing this game required me to put great effort.

I put much effort into this game.

I invested much energy into this game.

B.2. Chinese version of the SGDS

主观游戏难度量表 （以下列出了所有25 个问题，可根据需要使用。要对这个量表进行评分，您必须对"玩家胜任感"分量表的条目进行反向计分，方法是用8减去每个条目的回答分数，而其他条目则按常规方式计分。每个分量表的平均分越高，表示玩家认为该维度越困难。）

对于以下每个陈述，请使用下列尺度表明您对它们的同意程度：

1-非常不同意，2-不同意，3-有些不同意，4-中立，5-有些同意，6-同意，7-非常同意

游戏复杂度

我认为游戏要素之间的关系是复杂的。

我认为游戏给的信息太多了。

我认为游戏的规则是复杂的。

这个游戏很难理解。

游戏完成难度

这个游戏很难完成。

这个游戏看起来不可能赢。

这个游戏的目标是不可能达成的。

游戏游玩难度

玩游戏时我必须非常仔细地观察。

玩游戏时我必须仔细识别不同的事物。

快速思考是玩游戏的重要组成部分。

玩游戏时我必须迅速行动。

玩游戏时需要我的操作精确。

玩家胜任感

我感到自己有能力玩这个游戏。

我觉得自己在这个游戏中非常有能力和有效率。

我在这个游戏上很熟练。

我对自己在这个游戏中的表现感到满意。

我认为与其他玩家相比，我在这个游戏中表现得很好 。

我在这个游戏上超过了平均水准。

玩家压力感

玩游戏时我感觉非常紧张。

这个游戏的压力超出了我的极限。

我在玩游戏时感到了很大压力。

玩这个游戏让我十分灰心丧气。

玩家努力感

玩这个游戏需要我付出很大的努力。

我为这个游戏付出了很多努力。

我在游戏中投入了很多精力。

B.3. Japanese version of the SGDS

主観的ゲーム難易度尺度 (以下に25の質問がすべて列挙されており、必要に応じて使用することができます。このツールを採点に使用するには、「プレイヤー有能感」サブスケールにある質問のスコアを反転させる必要があります。つまり、質問のスコアを8から引きます。その他の部分は通常通り使用します。各サブスケールの平均スコアが高いほど、プレイヤーはその次元をより難しいと感じることになります。)

次のそれぞれの記述について、以下の尺度を使用してどの程度同意するかを示してください。

1-強く不同意、2-不同意、3-やや不同意、4-どちらでもない、5-やや同意、6-同意、7-強く同意

ゲームの複雑さ

このゲーム要素間の関係性が複雑だと思う。
このゲームは提供される情報が多すぎると思う。
このゲームのルールは複雑だと思う。
このゲームを理解するのはとても難しい。

ゲーム完了の難度

このゲームを完了するのは難しい。
このゲームで勝つことは不可能に見える。
このゲームの目標を達成することは不可能だ。

ゲームプレイの難度

ゲームプレイしているとき私は細心の注意を払わなければならなかった。
ゲームプレイ中私は様々なことを注意深く識別しなければならなかった。
素早く考えることはゲームをプレイする上で重要な部分であった。
ゲームプレイしているとき私は素早く動作しなければならなかった。
ゲームプレイでは動作に正確さが求められた。

プレイヤーの有能感

このゲームで私は有能だと感じる。
このゲームで私はとても有能で効率的だと感じる。
私はこのゲームに熟練している。
私はこのゲームでの自分のパフォーマンスに満足している。
他のプレイヤーと比較して、私はこのゲームでとても上手だったと思う。
このゲームで私は平均以上に優れている。

プレイヤーの重圧感

このゲームをプレイしている間、私はとても緊張していた。
このゲームで感じるストレスは私の限界を超えた。
このゲームをプレイしている間、私はとてもプレッシャーを感じた。
このゲームをプレイして私はとても気落ちした。

プレイヤーの努力感

このゲームをプレイするには多大な努力が求められた。
私はこのゲームに多くの努力を注いだ。
私はこのゲームに多くのエネルギーを注いだ。

Data availability

Data will be made available on request.

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